Review of Last Lecture

- 2.1 Introduction
- 2.2 Establishment of System Mathematical Model (Differential Equation)
- 2.3 Solving Differential Equations by the Classical Time-Domain Method
- 2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

2.2 Establishment of System Mathematical Model (Differential Equation)

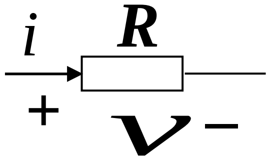
The goal of establishing a differential equation is to mathematically describe the physical characteristics of a system.

Electrical Circuit System Model

1) Component Constraints (Circuit Elements)

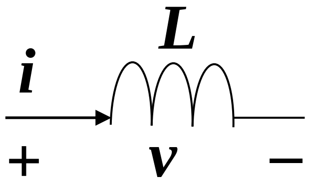
Voltage-Current Relationship

Resistor *R*



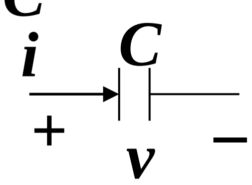
$$v = Ri$$

Inductor *L*



$$i = \frac{1}{L}\int v\,dt \qquad v = L\,di/dt$$

Capacitor *C*



$$i = C\frac{dv}{dt} \qquad v = \frac{1}{C}\int i\,dt$$

2) Network Topology Constraints

Kirchhoff's Current Law (KCL): At any instant, the sum of currents entering a node equals the sum of currents leaving the node:

$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Voltage Law (KVL): Around any closed loop, the sum of electromotive forces equals the sum of voltage drops across resistors:

$$\sum_k E_k = \sum_k U_k$$

2.3 Solving Differential Equations by the Classical Time-Domain Method

For complex systems, the relationship between the input x(t) and output y(t) is described by an ***n*-th order linear constant-coefficient ordinary differential equation (ODE)**:

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1}\frac{d^{n-1} y(t)}{dt^{n-1}} + \ldots + a_1 \frac{dy(t)}{dt} + a_0 y(t)$$
$$= b_m \frac{d^m x(t)}{dt^m} + b_{m-1}\frac{d^{m-1} x(t)}{dt^{m-1}} + \ldots + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

The **complete solution** of this ODE consists of two parts:

1) **Homogeneous solution**: Also called **free response** or **natural response**, it describes the system’s inherent behavior (determined by the system itself).
2) **Particular solution**: Also called **forced response**, it describes the system’s behavior under the influence of the input (determined by the input signal).

Step 1: Solve the Homogeneous Solution

The homogeneous solution satisfies the **homogeneous ODE** (set the input side to 0):

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \ldots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = 0$$

Characteristic Equation: $a_n \alpha^n + a_{n-1}\alpha^{n-1} + \ldots + a_1\alpha + a_0 = 0$

The roots $\alpha_1 \ldots \alpha_n$ are called the system's "natural frequencies".

Roots of the Characteristic Equation	Form of Homogeneous Solution (Coefficients $A_1, \ldots, A_n$ are undetermined)
Distinct real roots	$y_h(t) = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t} + \ldots + A_n e^{\alpha_n t}$
Repeated real roots	If α_1 is a *k*-th order repeated root, the corresponding term is $(A_1 t^{K-1} + A_2 t^{K-2} + \ldots + A_{K-1} t + A_K) e^{\alpha_1 t}$
Complex conjugate roots	If $\alpha_{1,2} = \alpha \pm j\beta$, the corresponding term is $e^{\alpha t}(A_1 \cos\beta t + A_2 \sin\beta t)$

Step 2: Solve the Particular Solution

The form of the particular solution depends on the input signal (or the "**free term**" of the ODE, i.e., the right-hand side of the ODE).

Free Term	**Form of Particular Solution $y_p(t)$**
E (constant)	B (constant)
t^p	$B_p t^p + B_{p-1} t^{p-1} + \ldots + B_1 t + B_0$
$e^{\alpha t}$	$\begin{cases} Be^{\alpha t} & \text{(If } \alpha \text{ is not a characteristic root.)} \\ Bt^k e^{\alpha t} & (k\text{: multiplicity of } \alpha) \end{cases}$
$\cos\omega_0 t / \sin\omega_0 t$	$B_1 \cos\omega_0 t + B_2 \sin\omega_0 t$

Step 3: Find Undetermined Coefficients

Substitute the complete solution $\boldsymbol{y(t) = y_h(t) + y_p(t)}$ into the original ODE, and solve for the undetermined coefficients A_1, A_2, A_n using the initial conditions (0_+ states).

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Definition of States

1. **0_- state**: The system state just before the input is applied (no input influence).
2. **0_+ state**: The system state just after the input is applied (used as initial conditions for solving the ODE, since **the ODE solution is valid for $t \geq 0_+$**.

Circuit Switching Rule

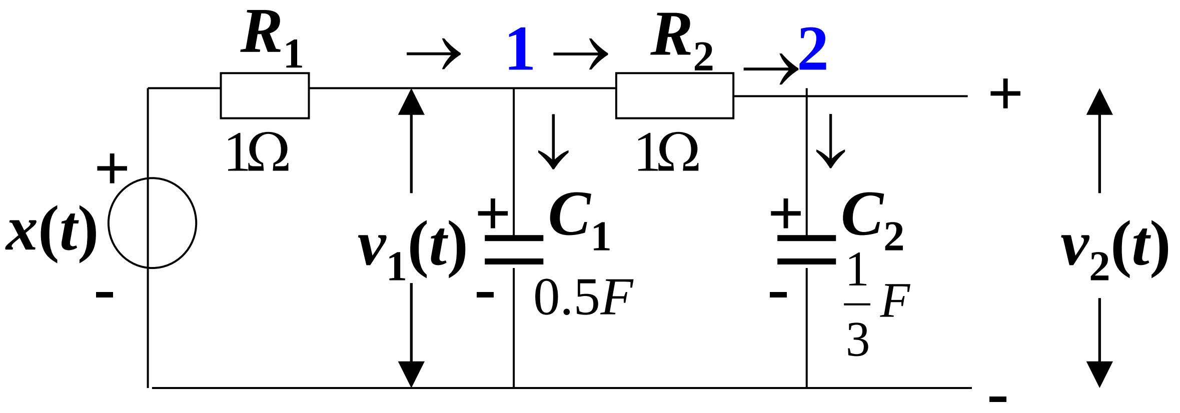
For circuit systems, the 0_- state describes the energy storage of components (capacitor voltage $v_C(0_-)$, inductor current $i_L(0_-)$.

In most cases, **the voltage across a capacitor and the current through an inductor do not change abruptly during switching**: $v_C(0_-) = v_C(0_+), \quad i_L(0_-) = i_L(0_+)$

However, **if an impulse current is applied to a capacitor or an impulse voltage to an inductor, $i_C(t)$ or $v_L(t)$ will jump between 0_- and 0_+**.

Example 2-3: For the given circuit (as shown in the figure), input $x(t)=\cos(2t)u(t)$, initial voltages of both capacitors are zero. Find the response $v_2(t)$.

Solution:



Step 1: Establish the ODE

List KCL equations for nodes 1 and 2,

$$\begin{cases} \dfrac{x(t)-v_1(t)}{R_1} = C_1\dfrac{dv_1(t)}{dt} + \dfrac{v_1(t)-v_2(t)}{R_2} & (1) \\[2ex] \dfrac{v_1(t)-v_2(t)}{R_2} = C_2\dfrac{dv_2(t)}{dt} & (2) \end{cases}$$

From (2), derive $$v_1(t)=R_2C_2\frac{dv_2(t)}{dt}+v_2(t) \quad (3)$$

Substituting (3) into (1), derive the ODE $$\frac{d^2v_2(t)}{dt^2}+7\frac{dv_2(t)}{dt}+6v_2(t)=6x(t)$$

ODE: $$\frac{d^2v_2(t)}{dt^2}+7\frac{dv_2(t)}{dt}+6v_2(t)=6x(t)$$

Step 2: Find the homogeneous solution

Characteristic equation: $\alpha^2+7\alpha+6=0$ Roots: $\alpha_1=-1,\alpha_2=-6$

Homogeneous solution: $A_1e^{-t}+A_2e^{-6t}$ (A_1 and A_2 are undetermined)

Step 3: Find the particular solution

Substitute x(t)=cos(2t) (t>0) into the right side of the ODE,

$$\frac{d^2v_2(t)}{dt^2}+7\frac{dv_2(t)}{dt}+6v_2(t)=6\cos 2t$$

so assume the particular solution is $B_1\sin 2t+B_2\cos 2t$. Substitute it into the left of the ODE,

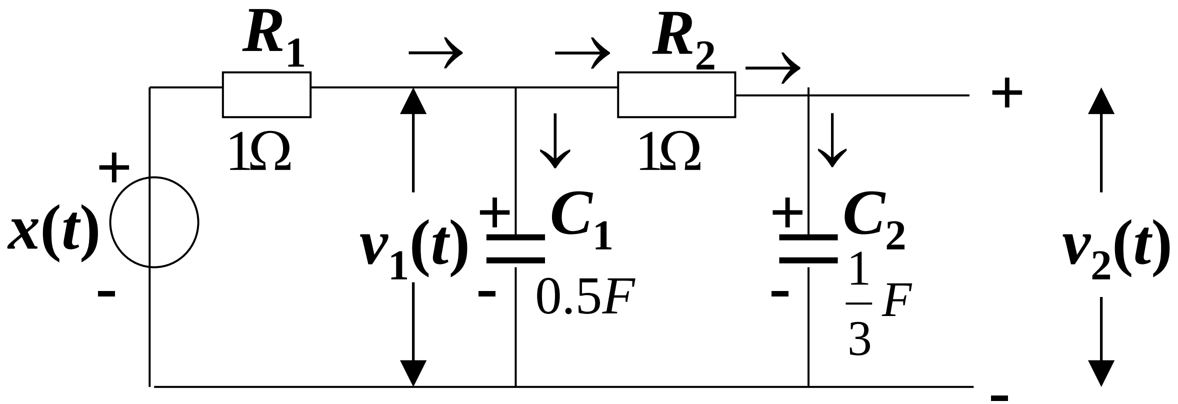
$$(2B_1-14B_2)\sin 2t+(14B_1+2B_2)\cos 2t=6\cos 2t$$

$$\begin{cases}2B_1-14B_2=0\\14B_1+2B_2=6\end{cases}\longrightarrow B_1=\frac{21}{50},B_2=\frac{3}{50}$$

Step 4: Find the complete solution

$$v_2(t) = A_1 e^{-t} + A_2 e^{-6t} + \frac{21}{50}\sin 2t + \frac{3}{50}\cos 2t \quad (t > 0)$$

$$\frac{dv_2(t)}{dt} = -A_1 e^{-t} - 6A_2 e^{-6t} + \frac{21}{25}\cos 2t - \frac{3}{25}\sin 2t$$



$x(t)=\cos(2t)u(t)$

As $v_2(0_-)=0$, and the voltage across C_2 does not change abruptly at t=0, hence $v_2(0_+)=0$; Similarly, $v_1(0_-)= v_1(0_+)=0$. The voltage across R_2 remains at 0 from 0_- to 0_+, and so there is no current through R_2 and C_2. Hence, $\frac{dv_2(0_+)}{dt} = 0$.

With $v_2(0_+)=0$, $\quad A_1 + A_2 = -\frac{3}{50}$

With $\frac{dv_2(0_+)}{dt} = 0$, $\quad A_1 + 6A_2 = \frac{21}{25}$

$\Rightarrow \quad A_1 = -\frac{6}{25}, A_2 = \frac{9}{50}$

Complete solution: $v_2(t) = -\frac{6}{25}e^{-t} + \frac{9}{50}e^{-6t} + \frac{21}{50}\sin(2t) + \frac{3}{50}\cos(2t) \quad (t > 0)$

Decomposition of the Complete Response:

Free/Natural Response (Homogeneous Solution): System's inherent behavior, from homogeneous ODE (input = 0), determined by natural frequencies and initial energy.

Forced Response (Particular Solution): System's behavior driven by input, form matches input (e.g., sinusoidal for sinusoidal input).

$$v_2(t) = \underbrace{-\frac{6}{25}e^{-t} + \frac{9}{50}e^{-6t}}_{\text{Free response (homogeneous solution)}} + \underbrace{\frac{21}{50}\sin(2t) + \frac{3}{50}\cos(2t)}_{\text{Forced response (particular solution)}}$$

Free response (homogeneous solution) **Forced response (particular solution)**

Decomposition of the Complete Response:

When the input signal is a step signal or a causal periodic signal, the complete response of a stable system can be divided into **transient response** and **steady-state response**. The **transient response** refers to the response that appears temporarily in the complete response after the excitation is applied, and gradually fades away as time increases. The **steady-state response** usually consists of step functions and periodic functions.

$$v_2(t) = \underbrace{-\frac{6}{25}e^{-t} + \frac{9}{50}e^{-6t}}_{\text{Transient response}} + \underbrace{\frac{21}{50}\sin(2t) + \frac{3}{50}\cos(2t)}_{\text{Steady-state response}}$$

Contents of This Lecture

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State (Continued)

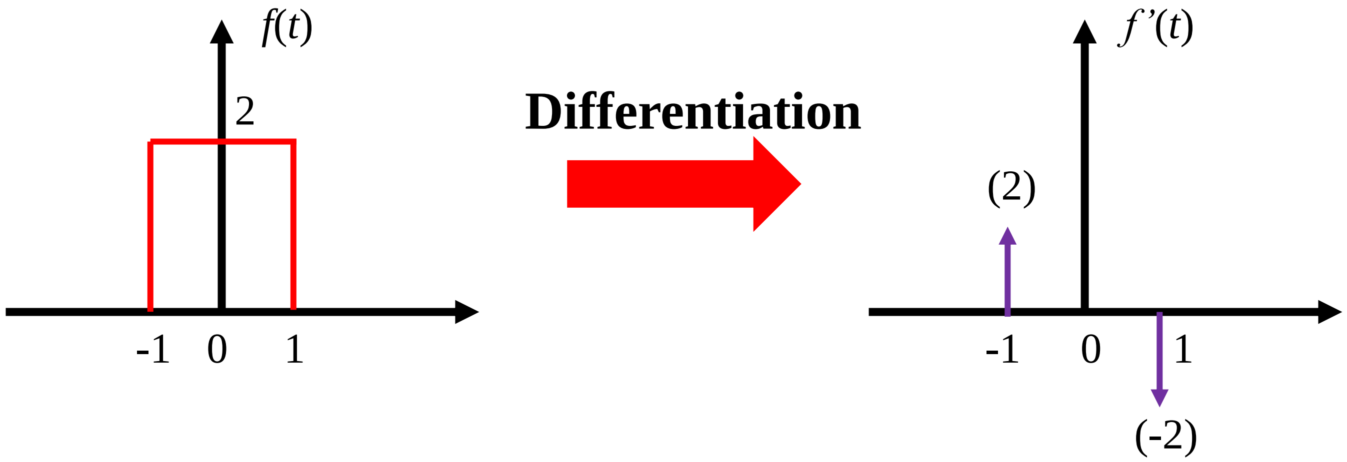
2.5 Zero-Input Response and Zero-State Response

Learning Objectives of This Lecture

1. Proficiently use the **impulse function matching method** to determine the jump variable of the starting point state;
2. Master the **classical method** to solve the **zero-input response** (homogeneous solution) of a system;
3. Proficiency in using the **classical method** and the **impulse function matching method** to solve the **zero-state response**;

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Impulse Function Matching Method: Impulse functions are the cause of state jumps (for example, an impulse current in a capacitor leads to a jump in its voltage). Therefore, **at t=0 , the δ(t) term and its derivatives of all orders on both sides of the differential equation must be balanced.**



$$f(t) = 2u(t+1) - 2u(t-1) \qquad \frac{df(t)}{dt} = 2\delta(t+1) - 2\delta(t-1)$$

If there is a state jump, its derivative must be an impulse function.

In general, the coefficients of the homogeneous solution can be solved by the impulse function matching method.

Example 2-4: Given $\frac{\mathrm{d}}{\mathrm{d}t}r(t)+3r(t)=3\delta'(t)$, and $r(0\_)=1$, find $r(0_+)$.

Solution: Since the right-hand side of the equation has $3\delta'(t)$, it must be part of $\frac{\mathrm{d}}{\mathrm{d}t}r(t)$.

Assume $\frac{\mathrm{d}}{\mathrm{d}t}r(t)=3\delta'(t)+a\delta(t)$

$r(t)=3\delta(t)+\boxed{a\Delta u(t)}$ **It means *r*(*t*) jumps by *a* from 0_- to 0_+.**

$\Delta u(t)$: **Relative Unit Jump Function** – State jump by 1 from 0_- to 0_+

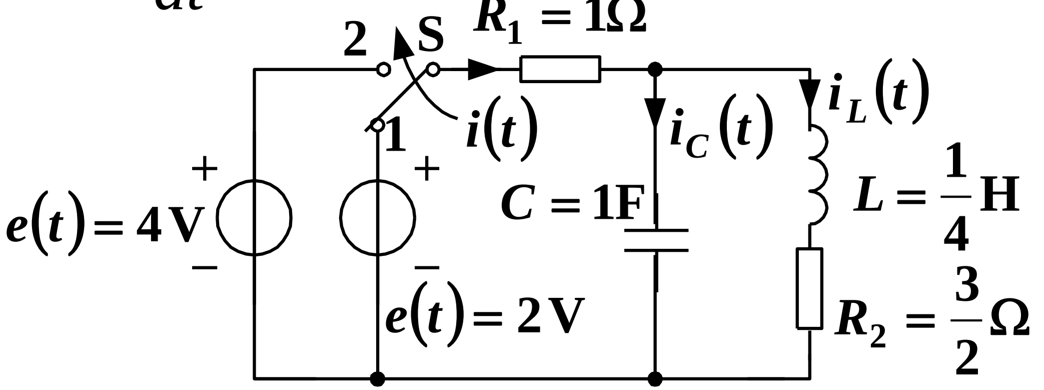
Substitute into the equation, and balance impulse terms

$3\delta'(t)+(a+9)\delta(t)=3\delta'(t)$ $\Rightarrow$ $a+9=0$ $\Rightarrow$ $a=-9$

$r(0_+)=r(0\_)+a=1-9=-8$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Use the Impulse Function Matching Method to find the state changes from 0_- to 0_+ in Example 2-2. Given the circuit shown in the figure, switch S is in position 1 and has reached a steady state. When t=0 , it switches from position 1 to position 2. Establish the differential equation for current i(t) and determine $i(0_+)$ and $\frac{d}{dt}i(0_+)$ after the switching.



Solution: The differential equation of the circuit:

$$\frac{d^2}{dt^2}i(t)+7\frac{d}{dt}i(t)+10i(t)=\frac{d^2}{dt^2}e(t)+6\frac{d}{dt}e(t)+4e(t)$$

At around $t=0_+$, $e(t)=e(0_-)+2\Delta u(t)=2+2\Delta u(t)$,

Substitute it, $\frac{d^2}{dt^2}i(t)+7\frac{d}{dt}i(t)+10i(t)=2\delta'(t)+12\delta(t)+8\Delta u(t)+8$

$$\frac{d^2}{dt^2}i(t)+7\frac{d}{dt}i(t)+10i(t)=2\delta'(t)+12\delta(t)+8\Delta u(t)+8$$

Assume $\frac{d^2}{dt^2}i(t)=2\delta'(t)+a\delta(t)$, then $\frac{d}{dt}i(t)=2\delta(t)+\underline{a}\Delta u(t)$, $i(t)=\underline{2}\Delta u(t)$

State jump amount from 0_- to 0_+

Balance the impulse functions and their derivatives,

$$2\delta'(t)+(a+14)\delta(t)=2\delta'(t)+12\delta(t)$$

$$a+14=12 \Rightarrow a=-2$$

$$\therefore i(0_+)=i(0_-)+2=0.8+2=2.8\text{ A},\quad \frac{d}{dt}i(0_+)=\frac{d}{dt}i(0_-)-2=0-2=-2\text{ A/s}$$

The coefficients of the homogeneous solution can then be solved with the states at 0_+.

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Classical Time-Domain Analysis of Circuit System

Pros: Clear physical meaning;
Cons: Complex process.

1. List circuit differential equations using Kirchhoff's Law

2. Convert differential equations into an input-output differential equation

3. Homogeneous solution (coefficients to be determined)

4. Look up the table to find a particular solution

5. Complete solution = homogeneous solution + particular solution. Find the coefficients of the homogeneous solution

Initial state 0_+

6. Complete solution--Total system response

2.5 Zero-Input Response and Zero-State Response

- The complete response of a system solved by the classical method can be divided into:

Complete Response = Free Response + Forced Response

$$r(t) = r_h(t) + r_p(t)$$

- The complete response of the system can also be divided into:

Complete Response = Zero-Input Response + Zero-State Response

$$r(t) = r_{zi}(t) + r_{zs}(t)$$

1. Definition of Zero-Input Response and Determination of Undetermined Coefficients

① Definition: The response generated entirely by the initial state without the action of an external excitation signal, i.e., $r_{zi}(t) = H[\{x(0_-)\}$

② Equation Satisfied：

$$c_0 \frac{d^n}{dt^n} r_{zi}(t) + \ldots + c_{n-1} \frac{d}{dt} r_{zi}(t) + c_n r_{zi}(t) = 0$$

Therefore, it is in the form of a homogeneous solution, i.e.,

$$r_{zi}(t) = \sum_{k=1}^{n} A_{zik} e^{\alpha_k t}$$

where $\alpha_{1,} \alpha_2 \ldots\ldots \alpha_n$ are n distinct system characteristic roots.

③ Initial Conditions：

$$r_{zi}^{(k)}(0_+) = r_{zi}^{(k)}(0_-) = r^{(k)}(0_-)$$

That is, the undetermined coefficients of the homogeneous solution $r_{zi}(t)$ can be determined by $r^{(k)}(0_-)$.

2. Definition of Zero-State Response and Determination of Undetermined Coefficients

① Definition: The response generated only by the excitation when the initial state is zero.

$$r_{zs}(t) = H[e(t)]$$

② Equation Satisfied: $c_0 \frac{d^n}{dt^n} r_{zs}(t) + \ldots\ldots + c_{n-1}\frac{d}{dt} r_{zs}(t) + c_n r_{zs}(t) = E_0 \frac{d^m}{dt^m} c(t) + \ldots.. + E_m$

Therefore, $r_{zs}(t) = \sum_{k=1}^{n} A_{zsk} e^{\alpha_k t} + r_p(t)$, i.e., it contains both the homogeneous solution and the particular solution.

③ Initial Conditions: Since the k-th ($k = 0, \ldots, n$-1) order derivatives of $r_{zs}(t)$ at time 0_- are all zero, i.e.,

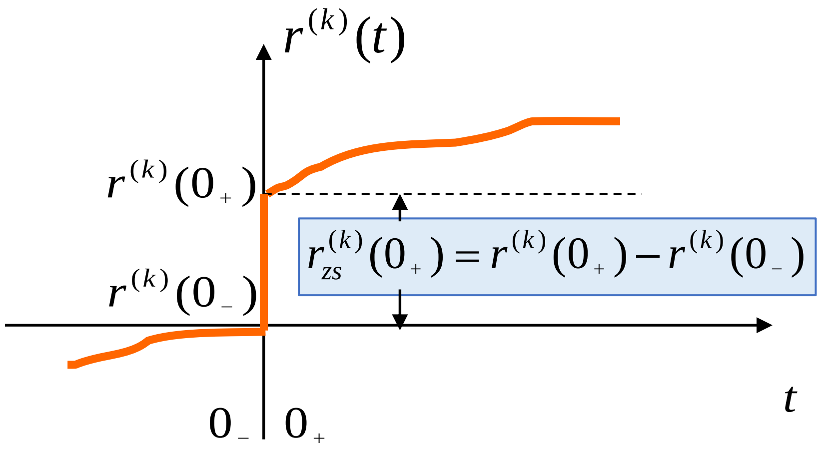
$$r^{(k)}(0_-) = r_{zi}^{(k)}(0_-) + r_{zs}^{(k)}(0_-) = r_{zi}^{(k)}(0_-) \qquad r_{zs}^{(k)}(0_-) = 0$$

$$r^{(k)}(0_+) = r_{zi}^{(k)}(0_+) + r_{zs}^{(k)}(0_+) = r_{zi}^{(k)}(0_-) + r_{zs}^{(k)}(0_+)$$

$$r_{zs}^{(k)}(0_+) = r^{(k)}(0_+) - r^{(k)}(0_-) = \text{Jump value}$$

Thus, A_{zsk} is determined by the jump value.

2.5 Zero-Input Response and Zero-State Response



$r^{(k)}(0_+)$ **:** Used to determine the coefficients of the **complete response**

$r^{(k)}(0_-)$ **:** Used to determine the coefficients of the **zero-input response**

$r_{zs}{}^{(k)}(0_+)$: Used to determine the coefficients of the **zero-state response**

Important!

$$r(t)=\underbrace{\sum_{k=1}^{n}A_{Zik}e^{a_k t}}_{\text{zero-input response}}+\underbrace{\sum_{k=1}^{n}A_{Zsk}e^{a_k t}+r_p(t)}_{\text{zero-state response}}$$

$$=\underbrace{\sum_{k=1}^{n}(A_{Zik}+A_{Zsk})\ e^{a_k t}}_{\text{free response}}+\underbrace{r_p(t)}_{\text{forced response}}$$

2.5 Zero-Input Response and Zero-State Response

Key Facts

- **Free Response**: Corresponding to the system characteristics;
- **Forced Response**: Corresponding to the excitation signal.
- **Zero-Input Response**: Caused by the internal energy storage of the system;
- **Zero-State Response**: Caused by the excitation signal.
- For many practical problems in communication and electronic systems, only the **zero-state response** needs to be calculated; to find the zero-state response, the cumbersome classical method is not required, and instead, the **convolution method** can be used.

Example 2-6: Given the differential equation of a system $r''(t)+3r'(t)+2r(t)=2e'(t)+6e(t)$, the initial state $r(0_-)=2$, $r'(0_-)=0$, and $e(t)=u(t)$, find the zero-input response and zero-state response of the system.

Solution: (1) Find the zero-input response $r_{zi}(t)$

$$r_{zi}''(t)+3r_{zi}'(t)+2r_{zi}(t)=0$$

Finding the initial value: $r_{zi}(0_+)=r_{zi}(0_-)=r(0_-)=2$

$$r_{zi}'(0_+)=r_{zi}'(0_-)=r'(0_-)=0$$

With the characteristic roots -1 and -2, assume $r_{zi}(t)=A_1e^{-t}+A_2e^{-2t}$

Substitute the initial values to find the coefficients $A_1=4$, $A_2=-2$

$$r_{zi}(t)=4e^{-t}-2e^{-2t} \quad (t>0)$$

Example 2-6: Given the differential equation of a system $r''(t)+3r'(t)+2r(t)=2e'(t)+6e(t)$, the initial state $r(0_-)=2$, $r'(0_-)=0$, and $e(t)=u(t)$, find the zero-input response and zero-state response of the system.

(2) Find the zero-state response $r_{zs}(t)$

$$r_{zs}''(t)+3r_{zs}'(t)+2r_{zs}(t)=2\delta(t)+6u(t)$$

$$r_{zs}(0_+)=0 \quad r_{zs}'(0_+)=r_{zs}'(0_-)+2=2$$

It is observed that $r(0)$ does not jump, while $r'(0)$ jumps by 2.

When $t>0$, $r_{zs}''(t)+3r_{zs}'(t)+2r_{zs}(t)=6$

Assume the homogeneous solution is: $r_{zsh}(t)=C_1e^{-t}+C_2e^{-2t}$

Assume the particular solution is a constant B. Substitute it into the equation: $2B=6$, so $B=3$.

Complete solution: $r_{zs}(t)=C_1e^{-t}+C_2e^{-2t}+3$ Its derivative: $r_{zs}'(t)=-C_1e^{-t}-2C_2e^{-2t}$

Substitute the initial states $r_{zs}(0_+)=0$ and $r_{zs}'(0_+)=2$, to find $C_1=-4$ and $C_2=1$.

Therefore, $r_{zs}(t)=-4e^{-t}+e^{-2t}+3 \quad (t>0)$

2.5 Zero-Input Response and Zero-State Response

Key Facts

Decomposing the response into zero-input response and zero-state response helps to understand the superposition and homogeneity characteristics of linear systems:

- **The zero-input response is linear with respect to the initial state** (initial energy storage) of the system;
- **The zero-state response satisfies linearity and time-invariance with respect to the excitation (input signal) of the system**.

Example 2-7: Given a linear time-invariant system, under the same initial conditions, when the excitation is $e(t)$, the complete response is $r_1(t)=[2e^{-3t}+\sin(2t)]u(t)$; when the excitation is $2e(t)$, the complete response is $r_2(t)=[e^{-3t}+2\sin(2t)]u(t)$.

(1) With the initial conditions unchanged, find the complete response $r_3(t)$ when the excitation is $e(t-t_0)$ $(t_0>0)$.

(2) When the initial conditions are doubled, find the complete response $r_4(t)$ when the excitation is $0.5e(t)$.

Solution: Let $r_{zi}(t)$ denote the zero-input response and $r_{zs}(t)$ the zero-state response.

Then, $\begin{cases} r_1(t)=r_{zi}(t)+r_{zs}(t)=\left[2e^{-3t}+\sin(2t)\right]u(t) \\ r_2(t)=r_{zi}(t)+2r_{zs}(t)=[e^{-3t}+2\sin(2t)]u(t) \end{cases} \Rightarrow \begin{cases} r_{zi}(t)=3e^{-3t}u(t) \\ r_{zs}(t)=[-e^{-3t}+\sin(2t)]u(t) \end{cases}$

(1) With the initial conditions unchanged, the complete response with the excitation $e(t-t_0)$ is

$$r_3(t)=r_{zi}(t)+r_{zs}(t-t_0)=3e^{-3t}u(t)+[-e^{-3(t-t_0)}+\sin(2t-2t_0)]u(t-t_0)$$

(2) When the initial conditions are doubled, the complete response with the excitation $0.5e(t)$ is

$$r_4(t) = 2r_{\mathrm{zi}}(t) + 0.5r_{\mathrm{zs}}(t)$$

$$= 2\left[3\mathrm{e}^{-3t}u(t)\right] + 0.5\left[-\mathrm{e}^{-3t} + \sin(2t)\right]u(t)$$

$$= \left[5.5\mathrm{e}^{-3t} + 0.5\sin(2t)\right]u(t)$$

Example 2-8: Suppose the differential equation of a linear time-invariant system is

$$\frac{d^2 r(t)}{dt^2}+3\frac{dr(t)}{dt}+2r(t)=\frac{de(t)}{dt}+3e(t)$$

where e(t) is the excitation and r(t) is the response. Given $e(t)=e^{-t}u(t)$, the complete response of the system is $r(t)=[(2t+3)e^{-t}-2e^{-2t}]u(t)$.

(1) Directly distinguish the forced response and the free response of the system, and give the explanation.

(2) Find the initial states $r(0\_)$ and $r^{'}(0\_)$ of the system using the impulse function matching method.

(3) Find the zero-input response and zero-state response of the system.

Solution:

(1) The free response is only related to the homogeneous solution.

The characteristic equation of the system is $\alpha^2 + 3\alpha + 2 = 0$

The characteristic roots are: $\alpha_1 = -1, \alpha_1 = -2$

Therefore, the free response (homogeneous solution) of the system is

$$r_h = (3e^{-t} - 2e^{-2t})u(t)$$

Subtracting the free response from the complete response gives the forced response as

$$\begin{aligned} r_p(t) &= r(t) - r_h(t) \\ &= [(2t+3)e^{-t} - 2e^{-2t}]u(t) - (3e^{-t} - 2e^{-2t})u(t) \\ &= 2te^{-t}u(t) \end{aligned}$$

(The excitation $e(t) = e^{-t}u(t)$ contains a characteristic root -1.)

(2) When $e(t)=e^{-t}u(t)$, $\dfrac{d^2r(t)}{dt^2}+3\dfrac{dr(t)}{dt}+2r(t)=e^{-t}\delta(t)-e^{-t}u(t)+3e^{-t}u(t)$

$$\frac{d^2r(t)}{dt^2}+3\frac{dr(t)}{dt}+2r(t)=\delta(t)+2e^{-t}u(t)$$

When t=0 , $\dfrac{d^2r(t)}{dt^2}+3\dfrac{dr(t)}{dt}+2r(t)=\delta(t)+2\Delta u(t)$

By the impulse function matching method, it is known that ***r'*(t) has a jump of +1 from 0_- to 0_+, while *r*(t) does not change from 0_- to 0_+**.

$$\frac{d^2r(t)}{dt^2}=\delta(t),\quad \frac{dr(t)}{dt}=1\cdot\Delta u(t),\ r(t)=0$$

$$\therefore r(0_+)=r(0_-),\quad r'(0_+)-r'(0_-)=1$$ **Jump amount of the state from 0_- to 0_+**

$$r(t)=[(2t+3)e^{-t}-2e^{-2t}]u(t),\quad r(0_+)=1$$

$$r'(t)=[2e^{-t}-(2t+3)e^{-t}+4e^{-2t}]u(t)=[-(2t+1)e^{-t}+4e^{-2t}]u(t),\quad r'(0_+)=3$$

$$r(0_-)=r(0_+)=1,\quad r'(0_-)=r'(0_+)-1=3-1=2$$

(3) Finding the zero-input and zero-state responses.

Method 1: Find the zero-input response first.

Characteristic roots of the system: $\alpha_1 = -1, \alpha_1 = -2$

Zero-input response: $r_{zi}(t) = A_1 e^{-t} + A_2 e^{-2t} \quad (t > 0)$

$$r_{zi}'(t) = -A_1 e^{-t} - 2A_2 e^{-2t} \quad (t > 0)$$

Substitute the initial states, $r_{zi}(0_+) = r_{zi}(0_-) = r(0_-) = 1,$

$$r_{zi}'(0_+) = r_{zi}'(0_-) = r'(0_-) = 2$$

$$\begin{cases} A_1 + A_2 = 1 \\ -A_1 - 2A_2 = 2 \end{cases} \Longrightarrow A_1 = 4, A_2 = -3$$

So the zero-input response is $r_{zi}(t) = (4e^{-t} - 3e^{-2t})u(t)$

The zero-state response is the complete response minus the zero-input response：

$$r_{zs}(t) = r(t) - r_{zi}(t) = [(2t-1)e^{-t} + e^{-2t}]u(t)$$

Method 2: Find the zero-state response first.

$$r_{zs}(t)=\underbrace{(A_{zs1}e^{-t}+A_{zs2}e^{-2t})u(t)}_{\text{homogeneous solution}}+\underbrace{Bte^{-t}u(t)}_{\text{particular solution}}$$

The particular solution is the forced response. It was solved in (1) as B=2. Alternatively, the particular solution and the excitation can be substituted into the left and right sides of the differential equation respectively to find B = 2.

It is known from (2) that that r(t) does not change from 0_- to 0_+, while r'(t) changes with an increment of 1.

$$\therefore r_{zs}(0_+)=r(0_+)-r(0_-)=0,\quad r'_{zs}(0_+)=r'(0_+)-r'(0_-)=1$$

$$r_{zs}(0_+)=A_{zs1}+A_{zs2}=0$$

$$r'_{zs}(0_+)=-A_{zs1}-2A_{zs2}+2=1$$

$$\rightarrow A_{zs1}=-1,\quad A_{zs2}=1.$$

$$r_{zs}(t)=(2te^{-t}-e^{-t}+e^{-2t})u(t)$$

The zero-input response is the complete response minus the zero-state response:

$$r_{zi}(t)=(4e^{-t}-3e^{-2t})u(t)$$

Homework

- Repeat the examples in the lecture notes.
- 2.55(a), 2.57(a)(c) and 2.58(a)(c) in the textbook.
- Additional questions on **the Impulse Function Matching Method**:

（1） $$2\frac{d^2r(t)}{dt^2}+3\frac{dr(t)}{dt}+4r(t)=\frac{de(t)}{dt}$$

Given $e(t)=u(t)$, $r(0\_)=1$, $r'(0\_)=1$, find $r(0_+)$ and $r'(0_+)$.

Solution: $r(0_+)=1, r'(0_+)=\frac{3}{2}$

（2） $$\frac{d^2r(t)}{dt^2}+5\frac{dr(t)}{dt}+6r(t)=\frac{de(t)}{dt}-2e(t)$$

Given $e(t)=u(t)$, $r(0\_)=1$, $r'(0\_)=2$, find $r(0_+)$ and $r'(0_+)$.

Solution: $r(0_+)=1, r'(0_+)=3$